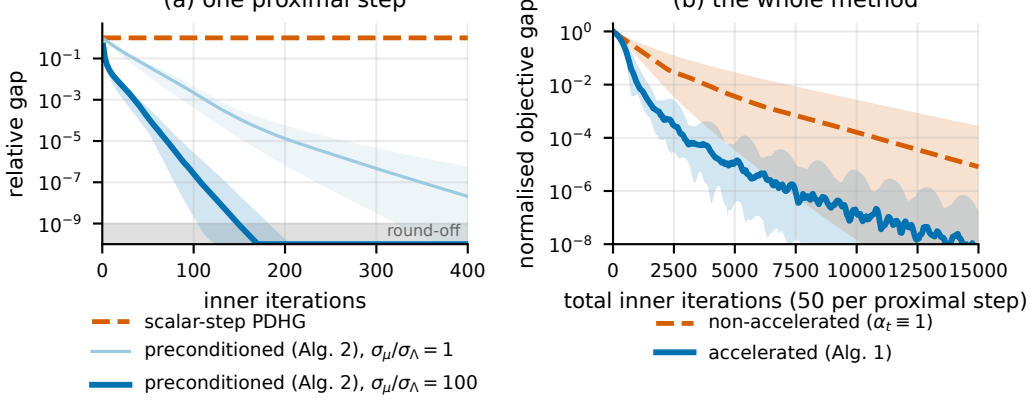
(a) one proximal step
relative gap
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
round-off
0
100
200
300
400
inner iterations
scalar-step PDHG
preconditioned (Alg. 2), $\sigma_\mu/\sigma_\Lambda = 1$
preconditioned (Alg. 2), $\sigma_\mu/\sigma_\Lambda = 100$
(b) the whole method
normalised objective gap
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
2500
5000
7500
10000
12500
15000
total inner iterations (50 per proximal step)
non-accelerated ($\alpha_t \equiv 1$)
accelerated (Alg. 1)